Where Does Intent Get Lost? An Offline Trajectory-Matching Analysis of Reasoning-to-Action Handoff Failures in a Humanoid VLA

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Abstract—Vision-Language-Action (VLA) models for humanoid manipulation typically decompose into a high-level reasoning stage that plans a sequence of sub-goals and a low-level action stage that executes continuous joint-space trajectories. Failures in these systems are often attributed vaguely to "the model," without distinguishing whether the plan itself was wrong or whether a correct plan failed to survive the handoff into execution. We present an offline evaluation methodology that scores these two stages independently against real teleoperated demonstrations and cross-references them into a joint classification, isolating the specific failure mode of intent lost in handoff: a correct plan whose execution diverges from the demonstrated trajectory. Applied to GR00T N1.7 (zero-shot on a real Unitree G1 teleop dataset) with Cosmos-Reason2-2B as a reasoning-stage proxy, across 50 pick-and-place episodes (150 sub-goal-phase observations) we find that 68.0% of phase observations show intent lost in handoff versus only 3.3% where both stages fail together, with failure concentrated in continuous joint control (arm, wrist, or torso deviates from the tightest tolerance in 100% of observations) rather than discrete grasp-state prediction (deviates in only 18.7%), and rising monotonically through the task from 58.0% at the initial reach to 74.0% by the final retreat. We discuss the methodological limits of trajectory-matching against a single demonstration as a proxy for task success, and release the scoring pipeline as an open, auditable tool for pre-deployment diagnosis of VLA planning-execution failures.

Index Terms—vision-language-action models, humanoid robots, failure analysis, offline evaluation, imitation learning

I. Introduction

Vision-language-action models increasingly follow a two-stage architecture explicitly described by their own authors in dual-process terms: a "System 2" component performs deliberate, language-mediated reasoning to decompose a task instruction into sub-goals, while a "System 1" component maps the current observation and sub-goal onto continuous low-level actions. GR00T N1 [1] and π_0 [2] both adopt this framing independently, and Embodied Chain-of-Thought [3] shows that eliciting an inspectable intermediate reasoning trace, even in an otherwise single-stage policy, improves both performance and interpretability. This decomposition is attractive for modularity, but it introduces a specific failure surface that single-stage, end-to-end policies [4], [5] do not have by construction: the reasoning stage can produce a completely correct plan that the action stage nonetheless fails to execute faithfully. We call this failure mode intent lost in handoff.

Existing evaluation practice for VLA models typically reports aggregate task success rate, which conflates planning failures with execution failures and gives no diagnostic signal about where in the reasoning-to-action pipeline a given failure originates. This is a meaningful gap for humanoid platforms specifically, where whole-body coordination (balance, torso posture, bimanual coordination) adds execution-stage failure modes beyond what tabletop single-arm manipulation exposes, and where physical trial-and-error evaluation is expensive and slow compared to software iteration.

We contribute:

1) An offline evaluation methodology that scores reasoning-stage and execution-stage fidelity independently, against ground truth derived directly from real teleoperated demonstrations (no simulation, no physical execution required), and cross-references them into a 2×2 classification that isolates intent-lost-in-handoff from cases where both stages fail together or where an incorrect plan is accidentally executed correctly.
2) A concrete instantiation of this methodology on GR00T N1.7 (NVIDIA's whole-body humanoid VLA policy) evaluated zero-shot against real Unitree G1 teleoperated demonstrations, using Cosmos-Reason2-2B as an explicit, labeled proxy for the reasoning stage GR00T does not expose in human-readable form.
3) An empirical failure taxonomy across 50 episodes (150 sub-goal-phase observations) showing that execution failures concentrate in continuous joint-space control rather than discrete grasp-state prediction, and that failure rate increases through the course of a pick-and-place task rather than being uniform.
4) Release of the full scoring pipeline as open, auditable tooling, including the specific numeric thresholds used and the reasoning behind design decisions made after inspecting real per-episode data (e.g., scoring

against an early prediction window rather than a full open-loop rollout, to isolate immediate plan-following from accumulated open-loop drift).

II. Related Work

VLA models and dual-system architectures. GR00T N1 [1], the model we evaluate, explicitly frames itself as a dual-system architecture: a vision-language module (its authors' "System 2") interpreting instructions and scene, coupled to a diffusion-transformer action module ("System 1") generating continuous motor commands. π_0 [2] independently converges on the same framing, pairing a discrete-token planning stage with a flow-matching action stage. Not every VLA makes this split explicit: OpenVLA [4] and RT-2 [5] are single-stage policies that map vision and language directly to action tokens without a separately inspectable planning representation. This distinction matters for our methodology, which requires a model (or a reasonable proxy for one) that produces an inspectable plan distinct from its low-level action output; it is not directly applicable to single-stage VLAs without first eliciting an analogous reasoning trace some other way.

Explicit reasoning and chain-of-thought in embodied models. Embodied Chain-of-Thought (ECoT) [3] trains OpenVLA to produce an inspectable reasoning trace – plans, sub-tasks, and grounded features like bounding boxes – before its action head runs, improving both success rate and interpretability. Our approach is related but serves a different purpose: ECoT modifies training to make an otherwise single-stage model produce visible reasoning as part of improving that same model, while we evaluate an already-dual-system, already-trained model's plan-to-action fidelity without any additional training, using reasoning and action as independently scored diagnostic signals rather than as a joint training target. Cosmos-Reason1 [6] – the predecessor to Cosmos-Reason2-2B, which we use as our reasoning-stage proxy (Section III) – is exactly the kind of standalone embodied-reasoning VLM this diagnostic approach depends on: a model that shares architectural lineage with GR00T's internal VLM but exposes natural-language chain-of-thought output that GR00T's own latent reasoning does not.

Failure analysis for robot policies. RoboFAC [7] is the closest prior work in spirit: a large-scale, failure-centric framework (9,440 erroneous trajectories, 78,623 QA pairs, simulation and real-world) that trains a dedicated model to diagnose and correct manipulation failures from trajectory data. It differs from our approach in two ways relevant to this paper's contribution: it categorizes failures without explicitly cross-referencing an independently-scored plan against independently-scored execution the way our 2×2 classification does, and it requires training a diagnostic model on a large labeled failure dataset, whereas our method is a lightweight, threshold-based offline rubric requiring no additional training data or model training, specifically designed to isolate the reasoning-to-action handoff as its own failure surface.

Evaluation methodology beyond success rate. Kress-Gazit et al. [8] argue that robot learning research over-relies on aggregate success rate reported with minimal experimental context, and propose richer practices: explicit conditions, multiple complementary metrics, statistical rigor, and qualitative failure description – demonstrated through physical hardware trials. We share this motivation but differ in setting: we obtain much of this richer diagnostic signal entirely offline against existing demonstration data, directly addressing the cost and slowness of physical trial-and-error evaluation that motivates their call for better practices in the first place.

Humanoid whole-body benchmarks and datasets. HumanoidBench [9] is a simulated benchmark spanning humanoid locomotion and manipulation tasks; Humanoid Everyday [10] is a large real-world dataset for diverse humanoid manipulation. Both establish task and data infrastructure for humanoid learning broadly; neither proposes a reasoning-versus-execution diagnostic methodology, which is the specific gap this paper targets. Like Humanoid Everyday, we use real (not simulated) demonstration data, but apply it to a narrower diagnostic question – where a specific model's plan-to-action fidelity breaks down – rather than broad task or skill coverage.

III. Method

A. Dataset

We use nvidia/PhysicalAI-Robotics-GR00T-Teleop-G1, a public dataset of real (not simulated) teleoperated Unitree G1 humanoid trajectories in LeRobot format, sampled at 20 fps. We evaluate on the g1-pick-apple task (pick up a red apple, place it on a plate). [TODO: note if additional task folders from this dataset – pear, grapes, starfruit – are added before submission for cross-task generalization.] State and action are 43-dimensional whole-body vectors (legs, waist, both arms, both hands).

B. Models

Action stage (System 1): GR00T N1.7 (3B parameters), a later checkpoint of NVIDIA's GR00T N1 whole-body humanoid VLA policy [1], loaded zero-shot with EmbodimentTag.REAL_G1 – a pretrain tag matched by construction to this dataset's embodiment, requiring no fine-tuning.

Reasoning stage (System 2) proxy: GR00T's own internal reasoning representation is latent (hidden-state activations), not human-readable. We use Cosmos-Reason2-2B, a vision-language model sharing backbone lineage with GR00T's internal VLM but released standalone with natural-language chain-of-thought output, run independently on the same task instruction and initial frame. Cosmos-Reason2 is the successor to Cosmos-Reason1 [6] and, as of this writing, is documented via model card and code release rather than a dedicated paper; we cite the Reason1 paper for the underlying architecture and training approach the Reason2 line descends from. This is an explicit proxy, not GR00T's literal internal state, and all results involving the reasoning stage must be read with this caveat.

C. Ground-truth sub-goal segmentation

The dataset provides one flat task instruction per episode, not labeled sub-goal boundaries. We derive reach / transport / retreat sub-goal boundaries from the recorded hand-joint trajectory: a scalar "hand closure" signal (mean absolute value of the 7 hand-joint state dimensions per frame, per hand) is computed for both hands; the hand with the larger observed range is taken as the grasping hand for that episode (this is computed from ground-truth recorded data, independent of any model prediction). A grasp event is the first sustained ($\geq$ 3-frame) crossing of a per-episode-fit threshold from below to above; a release event is the following sustained reverse crossing. This splits each episode into reach (start $\rightarrow$ grasp), transport (grasp $\rightarrow$ release), and retreat (release $\rightarrow$ end).

D. Execution-stage scoring

For each of the three sub-goal phases, GR00T is queried once at that phase's start frame, producing a predicted action chunk. This is compared against the actual recorded teleoperated action for the same window, per body-part group: wrist position (Euclidean distance, cm), wrist rotation (geodesic angle between predicted and actual rotation matrices, reconstructed from the 6D continuity representation of Zhou et al.), arm/torso joint angles (per-joint degree difference), and hand grasp state (discrete open/transitioning/closed bucket match). Table I gives the match/minor-deviation/failure thresholds per dimension.

TABLE I
Execution-stage scoring thresholds

Dimension	Match	Minor deviation	Failure
Wrist position	$< 3\,$cm	3–7 cm	$> 7\,$cm
Wrist rotation	$< 10°$	10–20°	$> 20°$
Arm/waist joint angle	$< 5°$	5–15°	$> 15°$

A body-part group (e.g., "left arm," 7 joints) is scored match if all joints are match-tier, failure if any single joint is failure-tier, minor deviation otherwise. Only the arm, hand, wrist, and torso of the grasping side (plus the shared torso/waist) drive the phase-level verdict; the idle arm's motion is not commanded by the task and its prediction error is not evidence of a failed handoff.

Scoring uses only the first 10 predicted steps ($\sim$0.5 s at this dataset's frame rate), not the full $\sim$40-step predicted chunk. Inspecting signed per-joint error on real data showed two distinct phenomena conflated by full-chunk averaging: some joints show error that grows steadily from near-zero over the open-loop horizon (expected behavior for any policy evaluated this way, since deployment normally re-plans every few steps rather than executing a long open-loop rollout), while others show a near-constant offset present even at the first predicted step (a more direct signal about whether the immediate action reflects the plan). We report both windows; the early window drives classification.

E. Reasoning-stage scoring

Cosmos-Reason2's step-by-step plan for the task is evaluated once per episode (it produces one plan up front, not a new plan per phase) via LLM-as-judge: does the plan cover all three canonical sub-goal phases, in the correct order, naming the correct object and target, without hallucinated content or being too vague to map onto the phases. We observed some inconsistency in the judge's handling of borderline elaborations across calls (Section V); this is a known limitation of LLM-as-judge approaches generally.

F. Joint classification

Per phase, we cross-reference reasoning-stage match against execution-stage tier (Table II). "Minor deviation" execution results are tracked as a separate category rather than folded into match or failure.

TABLE II
The core reasoning × execution classification

	Execution match	Execution failure
Reasoning match	Success	Intent lost in handoff
Reasoning failure	Accidentally correct	Compounding failure

IV. Results

Across 50 episodes (150 sub-goal-phase observations, all with valid grasp/release segmentation), Table III gives the overall classification breakdown and Table IV the breakdown by sub-goal phase.

TABLE III
Overall classification

Classification	Count	%
Intent lost in handoff	102	68.0%
Minor deviation	43	28.7%
Compounding failure	5	3.3%
Success	0	0.0%
Accidentally correct	0	0.0%

TABLE IV
Classification by sub-goal phase

Phase	Intent lost	Minor dev.	Compounding
Reach	58.0%	42.0%	0.0%
Transport	72.0%	26.0%	2.0%
Retreat	74.0%	18.0%	8.0%

Failure rate rises monotonically through the task: the model's execution is closest to the demonstrated trajectory immediately after the observation (reach), and diverges further as the task progresses through transport into retreat.

Table V shows which measured dimension most often drives a failure/minor-deviation verdict (a phase can implicate more than one dimension). Discrete grasp-state prediction (hand open/closed) is reliable; continuous joint-space and pose control (arm, wrist, torso) is where deviation concentrates. This suggests the model's difficulty is specifically in precise continuous control fidelity, not in higher-level object-interaction state tracking.

TABLE V
Which measured dimension drives failure/minor-deviation (of 150)

Dimension	Count
Waist (torso) rotation	148
Arm joint angles	145
Wrist position	145
Wrist rotation	100
Hand grasp state	28

Table VI gives reasoning-stage failure categories at the episode level. The dominant failure pattern is a correct plan with a failed execution (intent_lost_in_handoff, 68% of observations), not a failure of both stages together (compounding_failure, 3.3%) – consistent with the reasoning stage (or its proxy) being comparatively reliable relative to execution-stage trajectory fidelity in this setting. We manually verified all 3 compounding_failure episodes against their raw reasoning traces and execution numbers: in two, the plan never described a retreat/withdraw step at all; in the third, the plan described the plate moving to meet the apple rather than the reverse – a clear reversal of actor and object, not a borderline judge call.

TABLE VI
Reasoning-stage failure categories (episode-level, $n = 50$)

Category	Episodes
Hallucinated step	10 (20%)
Wrong object/target	9 (18%)
Missing sub-goal	5 (10%)

Table VII shows how many of the 5 scored dimensions (arm, hand, wrist position, wrist rotation, waist) land in the strictest match tier simultaneously, per phase. No phase in our sample achieves the strictest tier on all 5 dimensions simultaneously, which explains the 0% strict-success rate in Table III – but the shape of this distribution is itself informative. If a single overly strict threshold were responsible (e.g., the torso-rotation tolerance being unreasonably tight relative to what a well-performing model could achieve), we would expect phases to cluster near 4/5, with one holdout dimension. Instead the distribution is centered at 1–2/5 (82% of phases), with only 4 of 150 phases (2.7%) reaching 3 or more. This is consistent with genuinely independent, non-trivial deviation across multiple continuous-control dimensions at once, not an artifact of one miscalibrated cutoff.

TABLE VII
Dimensions matching simultaneously, per phase (of 5)

Dimensions matching	Count	%
0 of 5	23	15.3%
1 of 5	75	50.0%
2 of 5	48	32.0%
3 of 5	3	2.0%
4 of 5	1	0.7%
5 of 5	0	0.0%

V. Discussion and Limitations

Trajectory-matching is not task-success measurement. This is the central methodological caveat of the entire approach and must not be elided: every classification here measures divergence from one recorded human demonstration, not whether the predicted trajectory would have physically succeeded at the task. A pick-and-place task generally admits multiple valid solution paths; GR00T predicting a different-but-valid grasp approach would be scored identically to a genuinely failed grasp under this rubric. The zero success rate we observe (Table III) should be read as "GR00T's zero-shot predictions on this embodiment do not closely track this specific demonstrated trajectory," not as "GR00T fails to complete this task." Table VII's distribution supports this being a real measurement of partial trajectory fidelity rather than a rubric artifact – most phases achieve 1–2 of 5 dimensions within the strictest tolerance, not zero, and the shape of the distribution is inconsistent with one miscalibrated threshold being solely responsible. Grounding the trajectory-fidelity-vs-task-success distinction – ideally via the physical-consequence proxy layer described below – is the most important piece of future work this result motivates.

Reasoning-stage proxy. Cosmos-Reason2-2B is used as an explicit, labeled stand-in for GR00T's internal reasoning, which is otherwise a latent, non-human-readable representation. Results describing "reasoning-stage" behavior describe this proxy, not a literal decoding of GR00T's internals.

Forward-kinematics approximation. GR00T's REAL_G1 embodiment expects wrist end-effector pose (wrist_eef_9d) state/action fields that no public G1 dataset, including this one, records directly. We compute these via forward kinematics from the official Unitree G1 URDF. This is an approximation not independently verified against NVIDIA's internal training convention for this field; any bias in that convention should largely cancel out of the relative predicted-vs-actual comparisons this rubric performs (both sides are computed the same way), but this has not been independently confirmed.

LLM-as-judge noise. The reasoning-stage judge showed some inconsistency across calls in borderline cases – e.g., an unspecified color descriptor (“teal plate”) was judged harmless elaboration in one episode and flagged as hallucination in others. This is a known limitation of LLM-as-judge approaches generally rather than specific to this pipeline, and a source of noise in the reasoning-failure-category counts (Table VI) specifically, though it does not affect the execution-stage numbers (computed deterministically from numeric thresholds). We manually re-verified all 3 compounding_failure episodes – the cases most sensitive to judge reliability, since both stages must be independently correct for that label to be trustworthy – against the raw reasoning trace and execution numbers directly; all 3 held up as genuine double-failures rather than judge noise.

Single task, single object. Results here are from one task type (g1-pick-apple). [TODO: update if additional task folders are added before submission.] The dataset includes three other object types (pear, grapes, starfruit) under the same task structure; whether the phase-wise failure gradient and the discrete/continuous split generalize across object geometry is untested.

Sample size. 50 episodes / 150 phase observations is a modest sample for a strong generalization claim; error bars and statistical testing [TODO if time permits] would strengthen Tables III–VI’s claims, in the spirit of the statistical-rigor practices Kress-Gazit et al. [8] argue robot learning evaluation should adopt more broadly.

Physical-consequence proxy layer (explicitly out of scope here). A natural extension is a kinematic-plausibility layer estimating whether a flagged execution failure plausibly causes a physical consequence (e.g., a balance-relevant torso deviation, an unreachable target) – applied on top of, not in place of, the match/failure classification above. We deliberately scoped this out of the current submission to prioritize evaluating at a larger episode count within the available time; it remains the most direct way to address the trajectory-matching-vs-task-success gap above and is future work.

VI. Conclusion

[TODO: 1 paragraph restating the contribution and the headline empirical finding once final numbers are locked in.]

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